Clinical Evidence Summary: Protocol-Based Studies in Aerosol Delivery

Pediatric

This document provides an overview of peer-reviewed clinical studies evaluating aerosol drug delivery using vibrating mesh nebulizers (VMNs) across pediatric asthma care. The summaries highlight study design, results, and treatment protocols as described in the original publications. The data are presented for scientific and educational purposes only and are not intended to guide patient management, dosing, or device selection. Healthcare providers should refer to the full, published journal articles for complete methodology, clinical context, and results.

1. Moody et al., 2020[1]

FULL PUBLICATION

Design: Randomized controlled trial.

Population: 217 children (2–18 years) with moderate–severe asthma exacerbations (108 VMN, 109 JN).

Device assignment: JN used was the AirLife Sidestream High-Efficiency Nebulizer, CareFusion, Yorba Linda, California. VMN used was Aerogen Solo® with Aerogen Ultra® adapter, Aerogen, Galway, Ireland.

Results (as reported)

- **Median (IQR) number of treatments to reach a mild asthma score:** 2 (1–3) with VMN *vs* 3 (2-5) with JN (*P* < .001).
- Median (IQR) time to achieve a mild asthma score was 58 min with VMN *vs* 81 min with JN (*P* = .004).
- **Overall hospital admissions:** 15/108 (VMN) *vs* 22/109 (JN) (*P* = .22).
- In the subgroup treated with a face mask, VMN use was associated with a greater reduction in admission probability after adjustment (*P* = .032).

Protocol Snapshot – Children's Medical Center Acute Asthma Pathway

- **Scoring:** Asthma Score (AS) used to guide treatment and reassessment (mild: 1–4; moderate: 5–8; severe: 9–12).
- **First hour (AS 5–12):** 3 nebulized albuterol + ipratropium treatments every 20 minutes, with AS reassessment after each.
  - **If AS 1–4:** hold further doses; reassess every 20 min; if still mild after 40 min, evaluate for discharge.
  - **If AS 5–12:** continue next treatment.
- **Dosing:**
  - **<10 kg:** 2.5 mg albuterol / 250 µg ipratropium.
  - **≥10 kg:** 5 mg albuterol / 500 µg ipratropium.
- **Adjuvants (AS ≥9):** IV magnesium, terbutaline, or epinephrine per clinician.
- **Disposition:**
  - Mild (AS 1–4): discharge when SpO_2 ≥92% on room air, no distress, clinically stable.
  - AS 5–12: continue protocol; admit if not improving after 2 hours (6 treatments).

2. Crumm et al., 2025[2]

FULL PUBLICATION

Design: Retrospective observational study.

Population: 605 children (1-17 years; 302 pre-intervention, 303 post-intervention) receiving high-dose albuterol in a pediatric ED.

Device assignment: LVN (Hope Nebulizer, B&B Medical Technologies) and VMN (Aerogen Solo® with Aerogen Ultra® adapter, Aerogen, Galway, Ireland).

Results (as reported)

- **Total albuterol dose (adjusted mean):**
  - **Pre-intervention (conventional nebulizer):** 31.7 mg (95% CI: 30.2, 33.1).
  - **Post-intervention (VMN):** 15.9 mg (95% CI: 15.1, 16.1).
  - **Difference:** –15.8 mg (95% CI –17.4 to –14.2).
- **Time to disposition (adjusted mean):**
  - **Pre-intervention:** 305 min (95% CI: 293, 317).
  - **Post-intervention:** 269 min (95% CI: 259, 280).
  - **Difference:** –36 min (95% CI –51 to –20).
- **Hospital admission:** aOR* 1.1 (95% CI 0.7–1.7) – no significant difference.
- **ICU admission (among admitted):** aOR* 1.7 (95% CI 0.7–4.3) – no significant difference.
- **Change in respiratory score after first treatment:** adjusted difference –0.2 (95% CI –0.6 to 0.3) – similar between groups.
- **Safety outcomes:**
  - **Unplanned ICU transfer within 72 hr:** aOR* 0.4 (95% CI 0.1–1.6).
  - **ED return within 72 hr:** aOR* 0.7 (95% CI 0.3–1.9).
  - Neither difference was statistically significant.

Protocol Snapshot – ED High-Dose Nebulizer Pathway

- **Clinical pathway:** Standardized asthma/respiratory pathway used in both periods; treatment intensity guided by a respiratory score (0–12).
- **Pre-intervention (conventional nebulizer):**
  - **First hour (respiratory score >5):** 20 mg albuterol (20 mg/24 mL) + 1 mg ipratropium (1 mg/5 mL) delivered continuously over 1 hr.
  - **Subsequent hours:** Additional 20 mg albuterol over 1 hr as needed.
- **Post-intervention (VMN):**
  - **Trigger:** High-dose nebulizer initiated at respiratory score >6.
  - **First hour:** Total 7.5 mg albuterol + 0.5 mg ipratropium via VMN:
    - 2.5 mg albuterol + 0.5 mg ipratropium over 15–20 min (from a 2.5 mg/ 3mL albuterol solution; 0.5 mg/ 2.5 mL ipratropium solution), then
    - 5 mg albuterol (2.5 mg/3 mL albuterol solution) over 15–20 min (one refill of 6 mL reservoir).
  - **Subsequent hours:** 7.5 mg albuterol (7.5 mg/9 mL) over 30–40 min (one refill).
  - **O_2 flow:** 2 L/min per manufacturer recommendations.

*aOR = Adjusted Odds Ratio

3. Okay et al., 2025[3]

FULL PUBLICATION

Design: Double-blind randomized controlled trial.

Population: 80 children (7-15 years; MN group: 40; CN group: 40).

Results (as reported)

- **Spirometry:**
  - Both groups showed improvement in FEV_1, FVC, PEF, FEV_1/FVC, and FEF_{25-75} after treatment.
  - **FEV_1% change:** Mesh nebulizer (MN) 16.0% ± 13.7 *vs* compressor nebulizer (CN) 8.4% ± 8.6 (*P* = .001).
  - **FEF_{25-75}% change:** MN 54.7% ± 53.8 *vs* CN 32.9% ± 40.2 (*P* = .043).
- **Vital signs:**
  - Greater reduction in respiratory rate with MN (-1.18 ± 1.01 *vs* JN -0.58 ± 0.75 breaths/min; *P* = .002).
  - Increase in heart rate was higher with MN (2.13 ± 3.32 *vs* JN 0.88 ± 2.70 breaths/min; *P* = .022).
  - SpO_2 improved in both groups; between-group difference was not statistically significant (*P* = .391).
- **Follow-up and safety:**

– **48-hour readmission:** 10% (MN) *vs* 17.5% (CN); difference not statistically significant (*P* = .330).
  - Reported side effects (palpitations, flushing, tremor) occurred in both groups without significant differences.

Protocol Snapshot – Acute Asthma Treatment Trial

- **Medication:** Salbutamol 0.15 mg/kg per dose (maximum 5 mg).
- **Dosing schedule:** Three doses given every 20 minutes (total of 3 treatments per subject).
- **Device assignment:**
  - **Mesh nebulizer (MN):** Aerogen Solo®.
  - **Compressor nebulizer (CN):** Omron CompAir Pro NE-C900.
- **Assessments:**
  - Spirometry (FEV_1, FVC, FEV_1/FVC, PEF, FEF_{25-75}) performed before and after the 3-dose series.
  - Heart rate, respiratory rate, and SpO_2 recorded before and after treatment.
- **Blinding and environment:**
  - Investigators and assessing clinicians were blinded to device type.
  - Devices were concealed; treatments performed under standardized conditions with consistent equipment and infection-control procedures.

4. Andoh et al., 2025[4]

FULL PUBLICATION

Design: Quality improvement project.

Population: 7,276 pediatric ED encounters (ages 2–18 yrs 4,495 pre-intervention, 2,781 post-intervention).

Results (as reported)

- Average nebulized albuterol dose decreased from 17.42 mg to 11.57 mg, with reduction sustained for 16 months.
- ED length of stay for discharged patients decreased from 4.42 to 4.08 hours.
- No significant differences in overall ED LOS, admission rates, or 24-hour return visits.
- Post-intervention, 31% of patients received at least one treatment via a vibrating mesh nebulizer (VMN).

Device Identification Disclaimer:

The publication refers to a vibrating mesh nebulizer (VMN) but does not identify the brand or manufacturer of the device used.

Protocol Snapshot –

- Protocol followed the Asthma Clinical Score (ACS)–based tiered asthma management pathway.

First Hour:

- **Mild (ACS 2–4):** Metered-dose inhaler (MDI) with spacer only.
- **Moderate (ACS 5–7):** VMN recommended for three albuterol treatments every 20 minutes; reassessment before each dose; treatment held if ACS ≤2.
- **Severe (ACS 8–13):** VMN or jet nebulizer burst therapy if SpO_2 <92%; adjunct medications per clinician.

Second Hour:

- **ACS 3–5:** VMN recommended.
- **ACS ≥6:** VMN or jet nebulizer burst therapy.

Additional details:

- VMN not included in the order set for patients <2 years or for SpO_2 <92%.
- Emphasis on frequent reassessment and dose conservation.

References: 1. Moody GB, Ferguson CF, Davis C, et al. Respiratory Care. 2020;65(9):1331-1338. doi:10.4187/respcare.07238. **2.** Crumm P, et al. Pediatric Pulmonology. 2025;60(2):214-222. **3.** Okay O, et al. Respiratory Medicine. 2025; 237:107686. doi:10.1016/j.rmed.2025.107686. **4.** Andoh H, et al. Journal of Pediatric Health Care. 2025;39(1):65-73.

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